

Claire AI Full API Layer Specification

****CLAIRE AI -- FULL API LAYER SPECIFICATION****

Medium-Wide Layout Unicode Symbols Only Export-Ready

1. Overview

This API specification defines the complete interface for Claire AI's:

Deterministic Engine

Connected Intelligence Engine

Hybrid Engine

Breakthrough Engine

Portfolio Engine

Safety & Governance Layer

Mode-Switching Layer

This is the acquisition-critical artifact that shows Claire is a real, implementable system.

2. API Structure

Claire's API is divided into six domains:

1. ``/modes`` -- mode switching, mode status
2. ``/ingestion`` -- live data ingestion
3. ``/breakthroughs`` -- breakthrough detection & retrieval
4. ``/portfolio`` -- portfolio creation, retrieval, clustering
5. ``/safety`` -- safety events, red-line triggers, governance
6. ``/system`` -- health, logs, audit, metadata

3. API Endpoints

3.1 Mode API

****GET /modes/status****

Returns current mode and engine states.

****POST /modes/set****

Body:

mode: deterministic | connected | hybrid

reason: string

override: boolean

Triggers mode switch with safety checks.

****GET /modes/capabilities****

Returns capabilities available in current mode.

3.2 Ingestion API (Connected & Hybrid Only)

****POST /ingestion/tech****

Submit technology signals.

****POST /ingestion/market****

Submit market signals.

****POST /ingestion/competitor****

Submit competitor signals.

****POST /ingestion/regulatory****

Submit regulatory signals.

****GET /ingestion/status****

Returns ingestion pipeline health.

3.3 Breakthrough API

****POST /breakthroughs/detect****

Triggers breakthrough detection pipeline.

****GET /breakthroughs/{id}****

Retrieve breakthrough object.

****GET /breakthroughs/recent****

List recent breakthroughs.

****POST /breakthroughs/score****

Recalculate breakthrough scores.

3.4 Portfolio API

****POST /portfolio/create****

Creates a portfolio item from a validated breakthrough.

****GET /portfolio/{id}****

Retrieve portfolio item.

****GET /portfolio/clusters****

Retrieve portfolio clusters.

****GET /portfolio/themes****

Retrieve themes.

****POST /portfolio/promote****

Promote a breakthrough to a Portfolio Candidate Object (PCO).

****GET /portfolio/dashboard****

Retrieve dashboard-ready portfolio view.

3.5 Safety & Governance API

****GET /safety/events****

List safety events.

****POST /safety/trigger****

Manually trigger a safety event.

****GET /safety/matrix****

Retrieve safety trigger matrix.

****POST /governance/approve****

Approve a blocked breakthrough or portfolio item.

****POST /governance/block****

Block an item from promotion.

3.6 System API

****GET /system/health****

System health check.

****GET /system/logs****

Retrieve logs.

****GET /system/audit****

Retrieve audit trail.

****GET /system/metadata****

System metadata, version, mode, uptime.

4. API Authentication Model

Claire supports:

API keys

OAuth2 (enterprise)

Air-gapped token model (deterministic mode)

Modes enforce different auth levels.

5. API Error Model

Standardized error object:

```
{ "code": "string", "message": "string", "severity": "low | medium | high  
| critical", "mode": "deterministic | connected | hybrid", "timestamp":  
"ISO8601" }
```

6. API Rate & Safety Limits

(Not numerical -- conceptual)

Deterministic Mode: no external ingestion allowed

Connected Mode: ingestion governed by safety matrix

Hybrid Mode: dual-pipeline with safety enforcement

7. Summary

This API layer:

defines Claire's entire external interface

proves the system is implementable

satisfies acquisition-grade diligence

completes a major part of the missing 20-30%

****SECTION 1 OF 6 COMPLETE****